

Literature resolution of the Wigderson–Wigderson Fourier norm-ratio conjecture

Status note for run `fa_banach_001`

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Status: literature already answered. Huang–Liu–Wu explicitly identify Conjecture 4.13 of Wigderson–Wigderson and give its complete truth-by-exponent resolution.

Original conjecture

The source is Avi Wigderson and Yuval Wigderson, *The uncertainty principle: variations on a theme*, arXiv:2006.11206 [1]. In the current source PDF, Conjecture 4.13 appears on PDF page 33. For $q \in (1, \infty]$ and $0 \neq f \in \mathcal{S}(\mathbb{R})$, it defines

$$F_q(f) = \frac{\|f\|_q \|\widehat{f}\|_q}{\|f\|_2 \|\widehat{f}\|_2} = \frac{\|f\|_q \|\widehat{f}\|_q}{\|f\|_2^2}$$

and conjectures that, for every $q \neq 2$, the image of F_q is all of $\mathbb{R}_{>0}$. The source itself proves the endpoint $q = \infty$ in Theorem 4.12, but leaves all finite $q \neq 2$ as the conjecture.

Exact later answer

Linzhe Huang, Zhengwei Liu, and Jinsong Wu, *Quantum smooth uncertainty principles for von Neumann bi-algebras*, arXiv:2107.09057, explicitly say that they give a complete answer to this conjecture [2]. In the arXiv PDF, Section 4 begins on page 15, restates the source statement verbatim as “Conjecture 1 (Conjecture 4.13 in [22]),” and Theorem 4.3 on PDF page 16 proves:

1. If $1 < q < 2$ and $p = q/(q - 1)$, then every nonzero Schwartz function satisfies

$$F_q(f) \geq \left(\frac{p^{1/p}}{q^{1/q}} \right)^{1/2} > 0.$$

Thus the proposed image $\mathbb{R}_{>0}$ is impossible in this regime.

2. If $q > 2$, then the image of F_q is exactly $\mathbb{R}_{>0}$.

Together with the source’s $q = \infty$ result, this completely resolves the conjecture: it is true above 2 and false below 2. The authors knew they were answering the exact conjecture, rather than merely proving a theorem that implies it after an agent-supplied identification.

The answer paper was published as *Quantum Topology* 15 (2024), no. 3/4, 473–501, DOI 10.4171/QT/207.

Scope and remaining question

No part of Conjecture 4.13 remains open. Huang–Liu–Wu end the section with a new problem: for $1 < q < 2$, determine the optimal constant

$$C_q = \inf\{F_q(f) : 0 \neq f \in \mathcal{S}(\mathbb{R})\}.$$

That sharp-constant problem is distinct from the resolved image conjecture.

References

- [1] A. Wigderson and Y. Wigderson, *The uncertainty principle: variations on a theme*, Bull. Amer. Math. Soc. 58 (2021), 225–261; arXiv:2006.11206.
- [2] L. Huang, Z. Liu, and J. Wu, *Quantum smooth uncertainty principles for von Neumann bi-algebras*, Quantum Topol. 15 (2024), no. 3/4, 473–501; arXiv:2107.09057.